Shiqi (Stella) Chen

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Personal Profile

System and software engineer specializing in backend systems and platform infrastructure for healthcare applications serving 2M+ patients across 900+ hospitals and clinics. Broad technical foundation spanning distributed systems, machine learning, cryptography and security.

Experience

PocketHealth *Toronto, Canada*

Internship - Software Engineer (PEY) *May 2024 - Aug 2025*

- Implemented Prometheus metrics and internal alerting to monitor service health and traffic across backend systems.
- Built an automated pipeline using cron jobs to extract monthly Azure storage utilization metrics, streamlining cost tracking for the finance team.
- Implemented JWT authentication and gateway authorization for secure service-to-service communication in a distributed system.
- Designed and optimized dynamic GitLab CI/CD pipelines to modularize infrastructure workflows and significantly reduce deployment overhead.
- Built and maintained Kubernetes-based infrastructure on Azure, improving deployment automation and supporting containerized services, ingress controllers, and internal platform operations.
- Improved backend service security and reliability by addressing vulnerabilities identified through automated security scanning and infrastructure review processes.

Kua.ai *Zhuhai, China*

Internship - Software Engineer *June 2023 - Aug 2023*

- Reverse-engineered Midjourney's interface to build a seamless integration for Kua.ai, bridging the gap between raw generative AI capabilities and the high precision requirements for cross-border e-commerce product listing.
- Collaborated across the development team to overcome technical constraints and deliver a reliable end-user experience.

University of Toronto *Toronto, Canada*

Work Study - Research Assistant (Supervisor: Prof. Qun Wang) *Jan 2025 - Apr 2025*

- Built a model for the N-Vortex simulator to support simulation-based analysis and experimentation.
- Validated and refined the implementation through testing, debugging, and iterative improvement.

Work Study - Lecture Notes Assistant (Supervisor: Prof. Lisa Jeffrey) *Jan 2024 - Apr 2024*

- Prepared and refined lecture notes in elementary Lie theory.

Teaching Assistant - MAT135, MAT233, MAT102, MAT224, MAT223, MAT136, CSC236, CSC263, CSC347, CSC427 *Jan 2023 - Present*

- Led tutorials for math and CS courses. Coordinated tutorial logistics and developed course assessments to support course delivery.

Education

University of Toronto *Toronto, Canada*

Honours Bachelor of Science (CGPA: 3.95/4.00) *Sept 2021 - April 2026*

- Computer Science Specialist; Information Security Specialist; Mathematical Science Minor
- 2022 Summer - Dean's List Scholar, 2023 Winter - Dean's List Scholar, 2026 Winter - Dean's List Scholar

International Department, The Affiliated High School of South China Normal University *Guangzhou, China*

High school *Aug 2018 - June 2021*

Selected Coursework

MAT302 - Introduction to Algebraic Cryptography *Jan 2026 - Apr 2026*

- Studied algebraic and number-theoretic cryptography, focusing on public-key constructions and digital signatures, including Diffie-Hellman, ElGamal, RSA, and DSA/ECDSA.

CSC2125 - Topics in Software Engineering: Blockchain Technology & Engineering *Sept 2025 - Dec 2025*

- Modeled PBFT in TLA+ and used formal verification to analyze safety and fault tolerance in distributed consensus.
- Examined smart contract semantics and their dependence on protocol-level guarantees for secure decentralized execution.

CSC427 - Computer Security *Jan 2024 - Apr 2024*

- Investigated VM detection and anti-detection techniques in the broader context of virtualization security.
- Analyzed CVE-2015-5165 & CVE-2015-7504 as case studies in QEMU-based VM escape, using Debian as the guest environment.

Skills

Languages	Go, Python, Java, Kotlin, C/C++, SQL, TypeScript, UNIX/Shell, HTML/CSS, Solidity
Infrastructure & Tools	Skills, Git, Docker, Kubernetes, Terraform, GitLab CI, Prometheus, Microsoft Azure, LaTeX
Areas	Distributed Systems, Cryptography, Blockchain, Machine Learning, Computer Security, Infrastructure Automation